# Modeling Bimanual Perceptual Interactions in Macaque Monkeys

## Bayesian Hierarchical Modeling

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# Outline


Task and Data
Model A: Sessions Pooled
- Model Structure
- Synthetic Data
- Real Data

Model B: Sessions Hierarchical
- Model Structure
- Synthetic Data
- Real Data

Model C: Distractor Amplitude as a Covariate
- Model Structure
- Synthetic Data
- Real Data

Model Comparison and Recovery
- Predictive Performance
- Model Recovery: B vs A
- Model C vs Model B

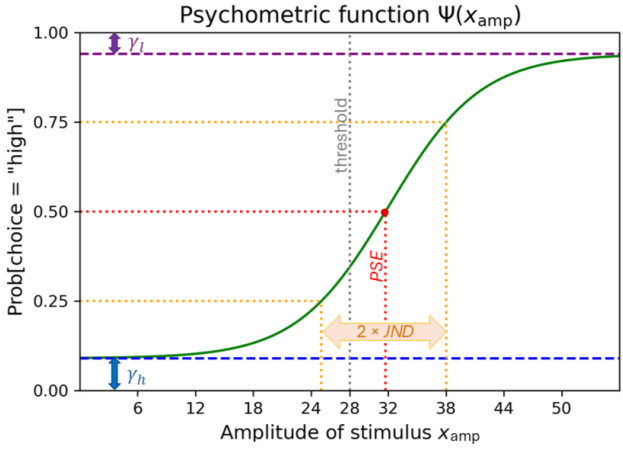
Summary

# The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\ \mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\ \mu m$
  - $x_{\text{dist}} \in [6, 32]\ \mu m$, value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 $\mu m$
  - "low" response: correct for $x_{\text{stim}} < 28\ \mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\ \mu m$
- correct responses are rewarded with water or juice

# Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

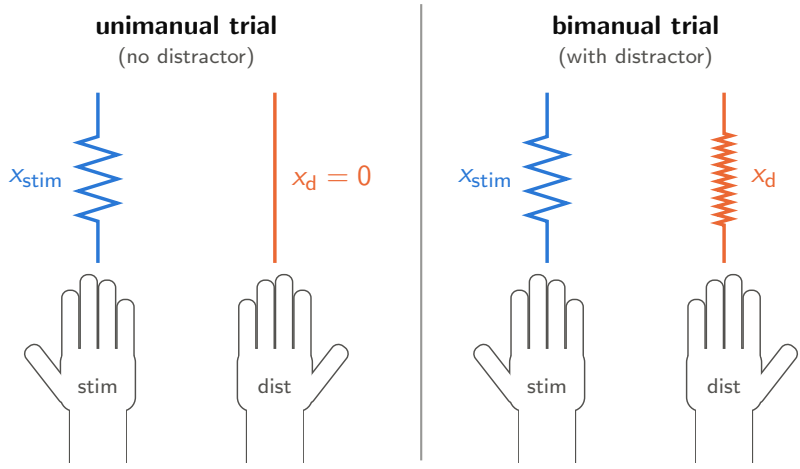


# Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

- $\gamma_h$: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of $\psi$
- $\gamma_l$: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of $\psi$
- $\text{logit}^{-1}$: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - $x_{\text{stim}}$ such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and $0.75$
  - represents sensitivity

# The task: one trial



# Modeling bimanual interactions

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\ \mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\ \mu m$
  - $x_{\text{dist}} \in [6, 32]\ \mu m$, value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 $\mu m$
  - "low" response: correct for $x_{\text{stim}} < 28\ \mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\ \mu m$
- correct responses are rewarded with water or juice

# The data

- One monkey (Sirius), 12,937 trials.
- Collected over 42 sessions, averaging approximately 308 trials per session.
- 8 stimulus amplitudes, 6–50 $\mu m$; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 $\mu m$) on its bimanual trials.

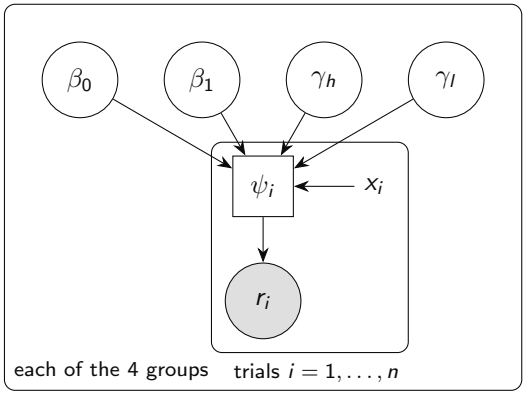
| | left hand | | right hand | | |
|---|---|---|---|---|---|
| $x_{\text{stim}}$ ($\mu m$) | unimanual | bimanual | unimanual | bimanual | total |
| 6 | 408 | 404 | 404 | 402 | 1,618 |
| 12 | 407 | 400 | 408 | 400 | 1,615 |
| 18 | 402 | 400 | 403 | 405 | 1,610 |
| 24 | 402 | 405 | 402 | 406 | 1,615 |
| 32 | 404 | 405 | 405 | 407 | 1,621 |
| 38 | 403 | 408 | 404 | 408 | 1,623 |
| 44 | 406 | 403 | 404 | 405 | 1,618 |
| 50 | 405 | 405 | 406 | 401 | 1,617 |
| total | 3,237 | 3,230 | 3,236 | 3,234 | 12,937 |

# Three models

| Model | Sessions | Distractor |
|---|---|---|
| A | pooled: one curve per group | uni and bi fit separately |
| B | hierarchical: one curve per group and session | uni and bi fit separately |
| C | hierarchical: one curve per hand and session | covariate: shift $\beta_2 x_{\text{dist}}$ |

For each model: convergence study on synthetic data, then the fit to the real data.

# Model A: graphical model




- circles: random variables
- shaded: observed
- rectangles: deterministic
- $x_i$: fixed covariate
- outer plate: the same model, fit independently to each group

# Model A: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials from all 42 sessions pooled:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta})]$ ($r_i = 1$ for "high", $x_i$ standardized), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta})^z \left(1 - \psi(x_i; \vec{\theta})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}] \\ &= \prod_{i=1}^{n} \psi(x_i; \vec{\theta})^{r_i} \left(1 - \psi(x_i; \vec{\theta})\right)^{1-r_i} \end{aligned}$$

# Model A: priors

The same priors in every group:

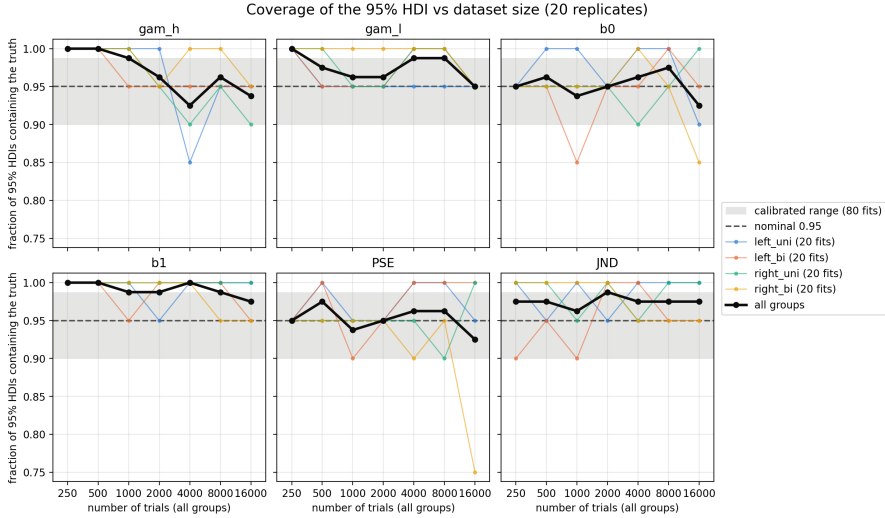
$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(0,\ 6) \\
\beta_1 &\sim \mathcal{N}(4,\ 2) \\
\gamma_h &= 0.25\, w_h, \qquad & w_h &\sim \text{Beta}(1.22,\ 4.89) \\
\gamma_l &= 0.25\, w_l, \qquad & w_l &\sim \text{Beta}(1.22,\ 4.89)
\end{aligned}$$

- $\beta_1$ should be positive for the psychometric function to have a positive slope
- $\gamma$'s have support between 0 and 0.25, with prior mean of 0.05 and 95% range $[0.003,\ 0.141]$

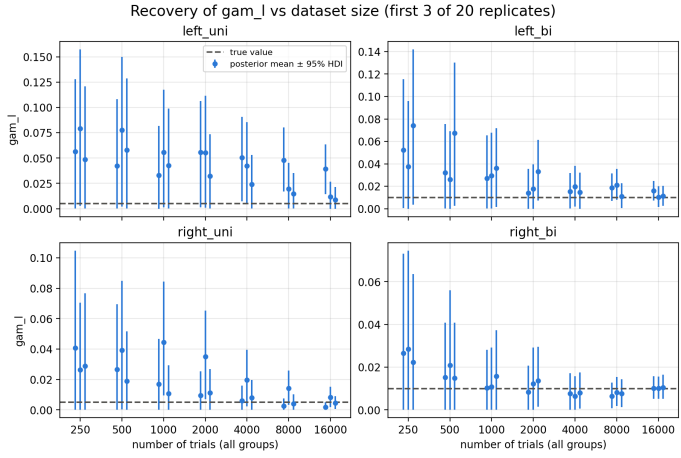
# Model A on synthetic data: the study

- One fixed set of true parameters $(\beta_0, \beta_1, \gamma_h, \gamma_l)$ per group
- Trials drawn from the real design (the real mix of stimulus levels and groups), responses simulated from the true curves
- 7 sizes: 250 to 16,000 trials; the real data (12,937) lie between the two largest
- 20 replicates of the whole sweep: 140 fits, 80 per size and parameter
- All fits sampled well: $\hat{R} \leq 1.006$, bulk ESS $\geq$ 1,554

# Model A on synthetic data: the 95% HDIs are calibrated



# Model A on synthetic data: $\gamma_l$ is pulled upward




True $\gamma_l$ (0.005, 0.01) sits in the prior's lower tail, and the data hold only 3–7 such lapses: the estimates are 1 posterior sd too high at small sizes, 0.36 sd at 16,000.

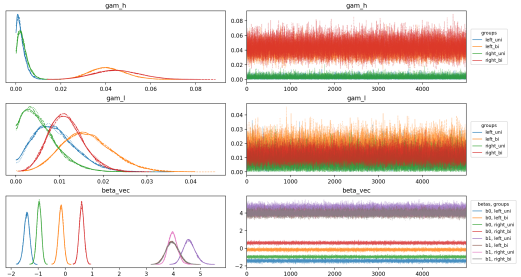
# Model A on synthetic data: what is recoverable

At the size of the real data (about 3,230 trials per group):

- **Recoverable:** PSE, JND, $\beta_0$, $\gamma_h$: unbiased, calibrated HDIs
  - posterior sd about 0.3 $\mu m$ for the PSE, 0.2 $\mu m$ for the JND
- **Recoverable, HDIs a little wide:** $\beta_1$ (coverage 0.98–0.99)
- **Not reliably recoverable:** small $\gamma_l$
  - biased upward by 0.4–0.5 sd; by about 1 sd when the curve never reaches its ceiling (left unimanual), where $\gamma_l$ trades off with the slope
  - HDIs still cover the truth, but the posterior mean should not be read literally
- **Caveats:** one set of true values; data simulated with one curve per group, so this says nothing about variation between sessions

# Model A on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (2.4 min)
- No divergences; tree depth never above 5
- All 40 parameters: $\hat{R} = 1.000$, bulk ESS $\geq$ 15,213, tail ESS $\geq$ 8,802



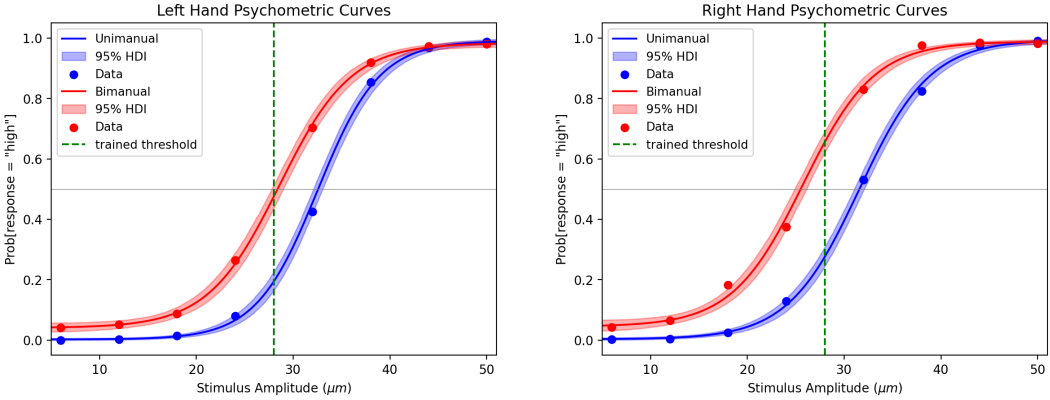
# Model A on the real data: group estimates

| | PSE ($\mu m$) | JND ($\mu m$) | $\gamma_h$ | $\gamma_l$ |
|---|---|---|---|---|
| left unimanual | 32.58 [32.10, 33.04] | 3.56 [3.25, 3.89] | 0.002 [0.000, 0.006] | 0.009 [0.000, 0.018] |
| left bimanual | 28.38 [27.79, 28.98] | 4.39 [3.93, 4.83] | 0.041 [0.025, 0.056] | 0.016 [0.005, 0.027] |
| right unimanual | 31.57 [31.05, 32.06] | 4.09 [3.77, 4.41] | 0.003 [0.000, 0.008] | 0.006 [0.000, 0.013] |
| right bimanual | 25.50 [24.89, 26.08] | 4.36 [3.93, 4.81] | 0.045 [0.025, 0.064] | 0.012 [0.004, 0.020] |

Posterior mean [95% HDI], 12,937 trials.

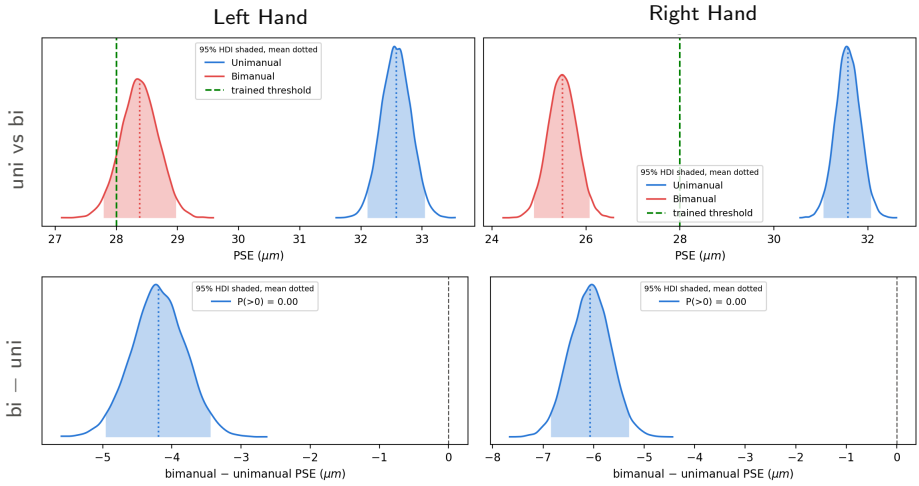
- Without the distractor the PSE lies 3.6–4.6 $\mu m$ above the trained 28 $\mu m$
- With it the PSE moves to the threshold (left) or below it (right)
- $\gamma_l$ estimates lean high (see the synthetic study)

# Model A on the real data: fitted curves



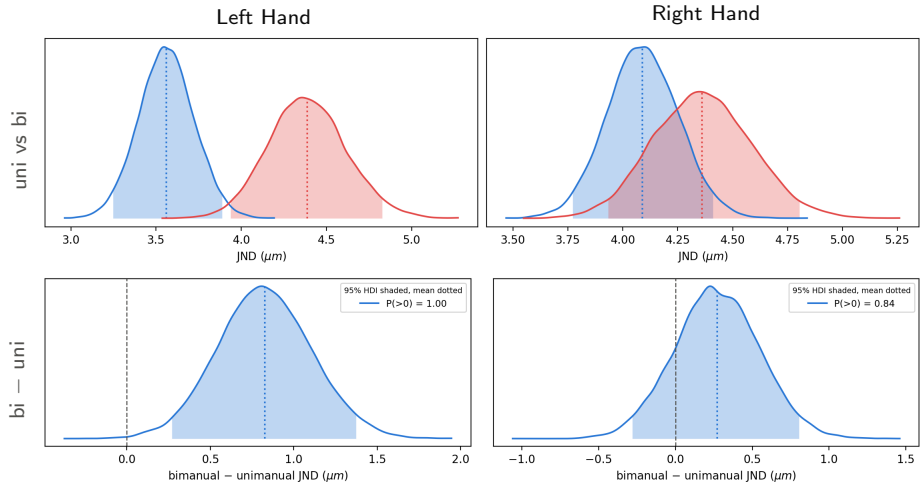

Unimanual in blue, bimanual in red, with the observed proportions pooled over sessions; dashed: the trained threshold of 28 $\mu m$.

# Model A on the real data: effect of the distractor on PSE

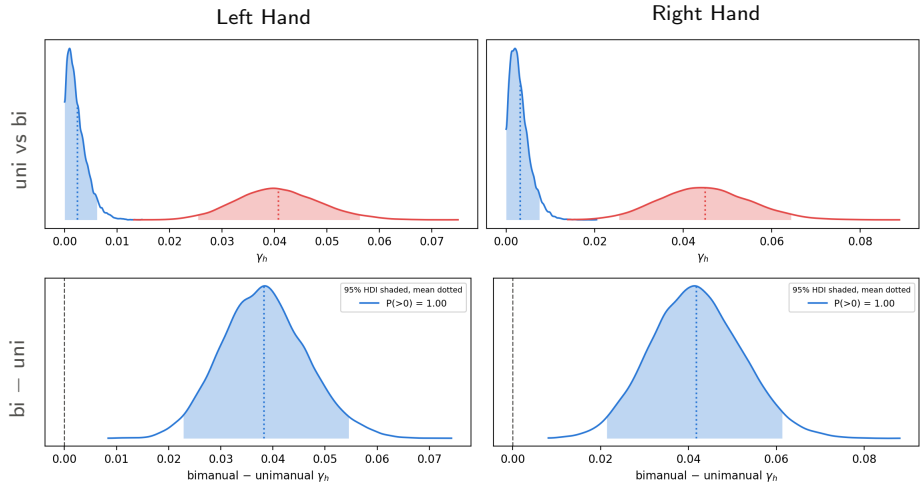



Unimanual Bimanual 95% HDI shaded, mean dotted; green dashed: trained threshold

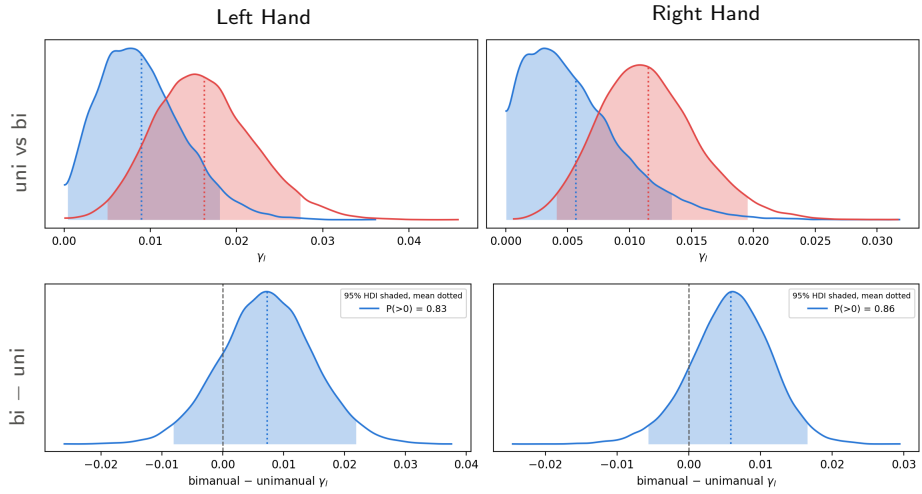
# Model A on the real data: effect of the distractor on JND



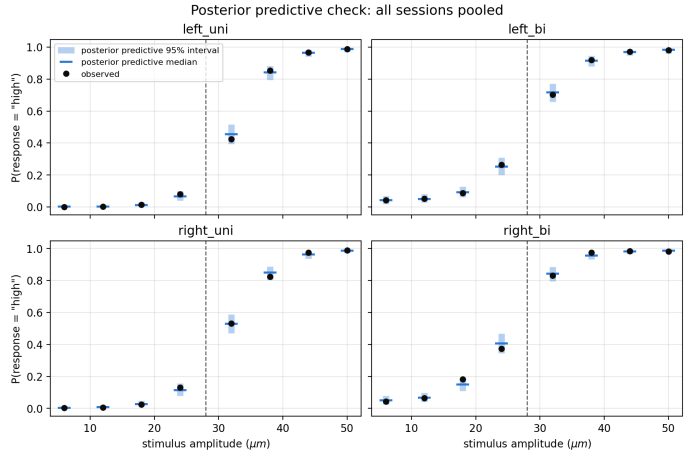
# Model A on the real data: effect of the distractor on $\gamma_h$



# Model A on the real data: effect of the distractor on $\gamma_l$



# Model A on the real data: the pooled data are reproduced



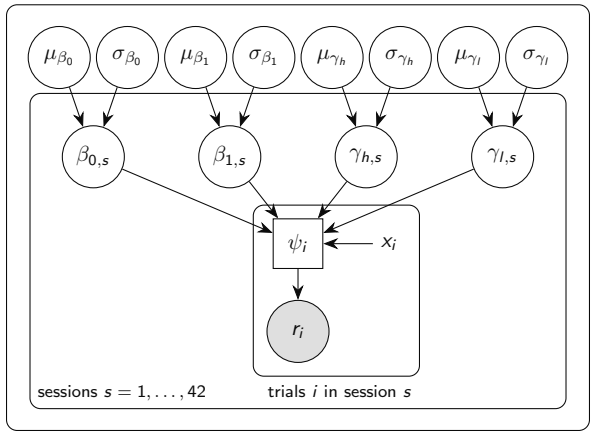

Pooled over sessions, all 32 group $\times$ amplitude proportions lie inside their 95% posterior predictive intervals.

# Model A: conclusions

- Reliable fit; the pooled data are reproduced at every amplitude ($\text{elpd}_{\text{loo}} = -3344.0$, SE 64.5)
- Effect of the distractor (bimanual − unimanual), posterior mean [95% HDI]:
  - **PSE:** lowered, by $-4.19$ [$-4.96$, $-3.44$] $\mu m$ (left) and $-6.07$ [$-6.85$, $-5.30$] $\mu m$ (right)
  - **JND:** raised for the left hand, $+0.83$ [0.27, 1.37] $\mu m$; unclear for the right, $+0.27$ [$-0.29$, 0.80] $\mu m$
  - $\gamma_h$**:** raised, by $+0.038$ [0.023, 0.055] (left) and $+0.042$ [0.021, 0.061] (right)
  - $\gamma_l$**:** no clear effect, $+0.007$ [$-0.008$, 0.022] (left) and $+0.006$ [$-0.006$, 0.017] (right)

$\Rightarrow$ A curve for every session: Model B

# Model B: graphical model




Circles: random variables (shaded: observed); rectangle: deterministic; $x_i$: fixed covariate. The same model is fit independently to each group.

# Model B: likelihood

For each group (left/right $\times$ unimanual/bimanual), fit independently to its $n \approx 3{,}230$ trials, now with a parameter set for every session $s$:

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\vec{\theta}_s)_{s=1}^{42}$,
$\vec{\theta}_s = (\beta_{0,s}, \beta_{1,s}, \gamma_{h,s}, \gamma_{l,s})$
$r_i \sim \textbf{Bernoulli}[\psi(x_i; \vec{\theta}_{s[i]})]$ ($s[i]$: the session of trial $i$), so:

$$\mathbf{P}[r_i = z] = \psi(x_i; \vec{\theta}_{s[i]})^z \left(1 - \psi(x_i; \vec{\theta}_{s[i]})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}_{s[i]}] \\ &= \prod_{s=1}^{42} \prod_{i:\, s[i]=s} \psi(x_i; \vec{\theta}_s)^{r_i} \left(1 - \psi(x_i; \vec{\theta}_s)\right)^{1-r_i} \end{aligned}$$

# Model B: priors

Session parameters drawn from the group's population (non-centered in the code):

$$\beta_{0,s} \sim \mathcal{N}(\mu_{\beta_0},\ \sigma_{\beta_0}), \qquad \gamma_{h,s} = 0.25\, w_{h,s}, \quad w_{h,s} \sim \text{LogitNormal}(\mu_{\gamma_h},\ \sigma_{\gamma_h})$$
$$\beta_{1,s} \sim \mathcal{N}(\mu_{\beta_1},\ \sigma_{\beta_1}), \qquad \gamma_{l,s} = 0.25\, w_{l,s}, \quad w_{l,s} \sim \text{LogitNormal}(\mu_{\gamma_l},\ \sigma_{\gamma_l})$$

Hyperpriors, the same in every group:

$$\mu_{\beta_0} \sim \mathcal{N}(0,\ 2), \qquad \sigma_{\beta_0} \sim \text{HalfNormal}(2),$$
$$\mu_{\beta_1} \sim \text{LogNormal}(\log 4,\ 0.5), \qquad \sigma_{\beta_1} \sim \text{Exponential}(1),$$
$$\mu_{\gamma_h},\ \mu_{\gamma_l} \sim \mathcal{N}(-3,\ 1.5), \qquad \sigma_{\gamma_h},\ \sigma_{\gamma_l} \sim \text{HalfNormal}(0.5)$$

- Chosen from the design, checked by prior predictive simulation, not fit to the data
- $\mu_{\beta_0}$: typical PSE within 13–43 $\mu m$ with 90% probability
- $\mu_{\beta_1} > 0$: median 4, 95% in [1.5, 10.6]; $\sigma_{\beta_1}$ keeps 98% of session slopes positive
- Typical lapse rate: median 0.012, 95% in [0.0007, 0.12]

# Model B on synthetic data: the study

- True hyperparameters = posterior means of Model B's fit to the real data (next slide), so the synthetic data resemble the real data; session parameters drawn from them
- Two sweeps, each run twice with new data:
  - sessions: 5, 10, 20, 40, 80 sessions of 308 trials (tests the hyperparameters)
  - trials: 42 sessions of 20, 40, 80, 160, 320 trials (tests the session parameters)
- The real data (42 sessions of about 308 trials) are closest to the 40-session and 320-trial fits
- All 20 fits sampled well: no divergences, $\hat{R} \leq 1.011$, bulk ESS $\geq 670$

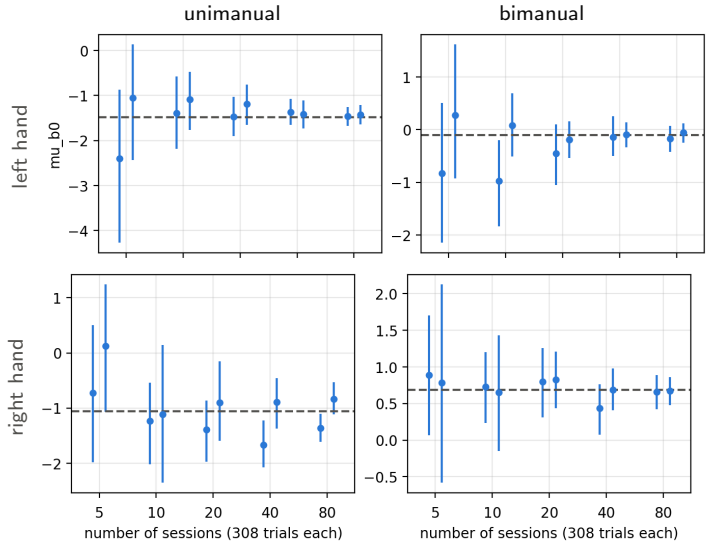
# Model B on synthetic data: the true values

Posterior means of the real-data fit (lapse-rate hyperparameters on the logit scale):

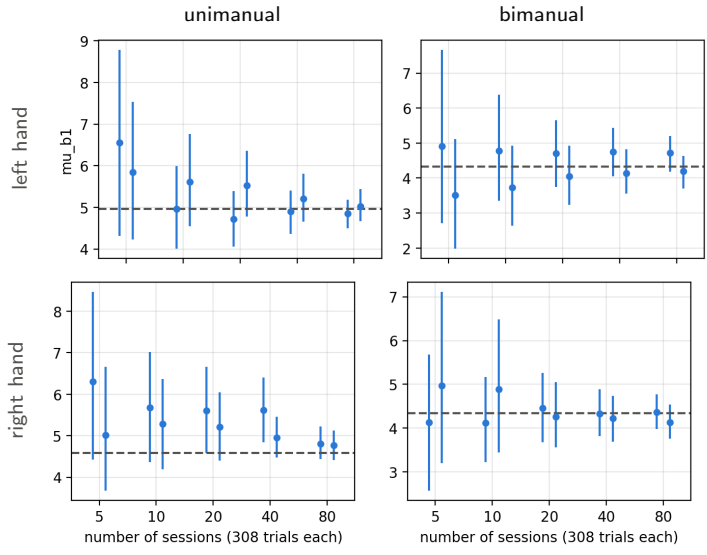
| | $\mu_{\beta_0}$ | $\mu_{\beta_1}$ | $\mu_{\gamma_h}$ | $\mu_{\gamma_l}$ | $\sigma_{\beta_0}$ | $\sigma_{\beta_1}$ | $\sigma_{\gamma_h}$ | $\sigma_{\gamma_l}$ |
|---|---|---|---|---|---|---|---|---|
| left unimanual | −1.49 | 4.97 | −5.20 | −3.44 | 0.76 | 0.29 | 0.36 | 0.37 |
| left bimanual | −0.10 | 4.32 | −2.35 | −3.44 | 0.77 | 1.27 | 0.59 | 0.39 |
| right unimanual | −1.06 | 4.58 | −4.73 | −3.72 | 1.01 | 0.17 | 0.37 | 0.38 |
| right bimanual | 0.69 | 4.34 | −1.82 | −3.23 | 0.76 | 0.62 | 0.36 | 0.43 |

- Group slopes 4.3–5.0, and slopes that vary between sessions (left bimanual most)
- Typical $\gamma_h$ near 0 on unimanual trials (0.001–0.002), 0.02–0.04 on bimanual
- Replaces an earlier study with hand-picked values (slopes 3 and 6, sessions with almost equal slopes and lapse rates) that did not resemble the real data

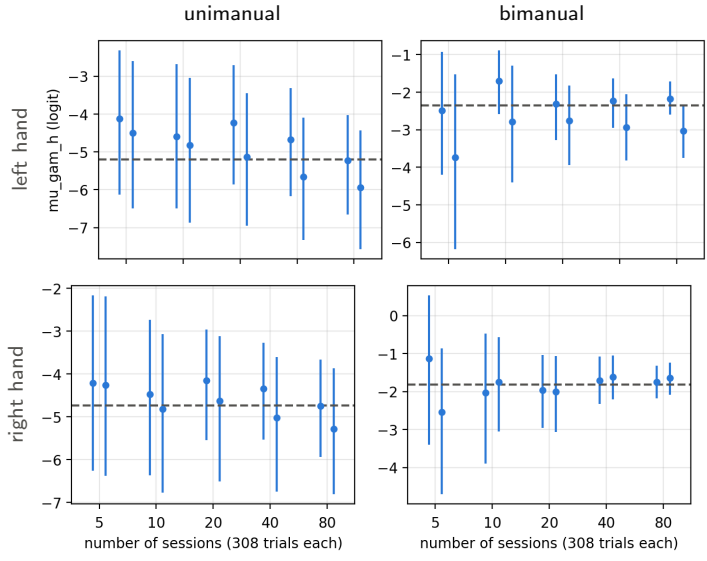
# Model B on synthetic data: $\mu_{\beta_0}$ vs number of sessions



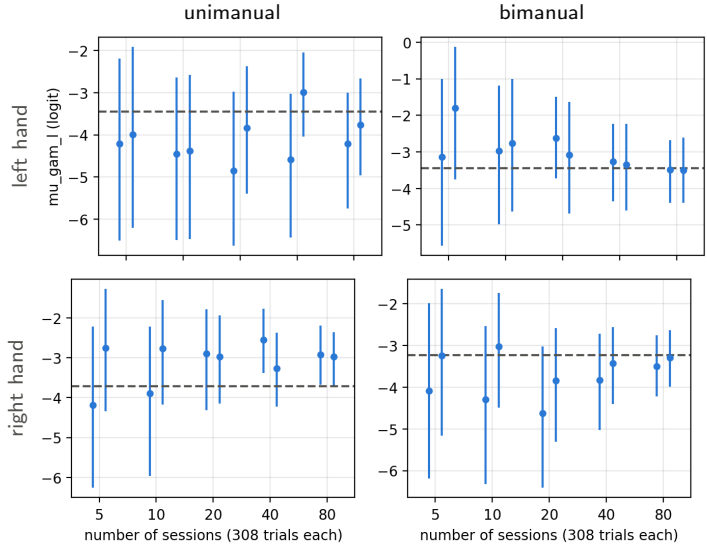
# Model B on synthetic data: $\mu_{\beta_1}$ vs number of sessions



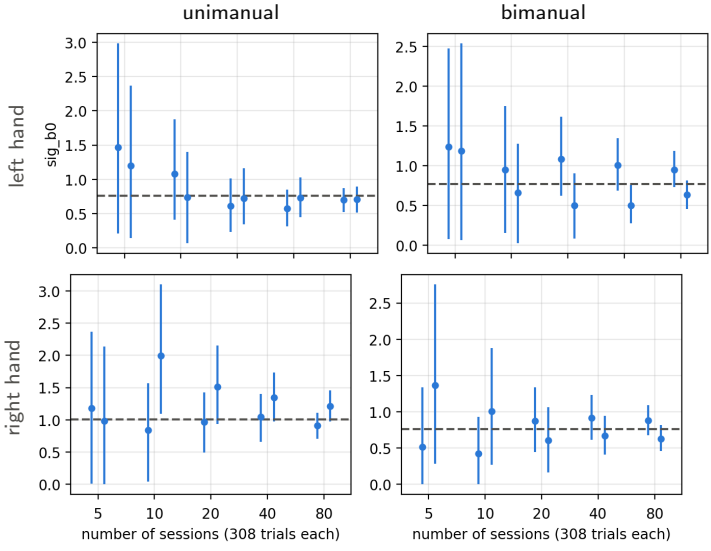
# Model B on synthetic data: $\mu_{\gamma_h}$ vs number of sessions



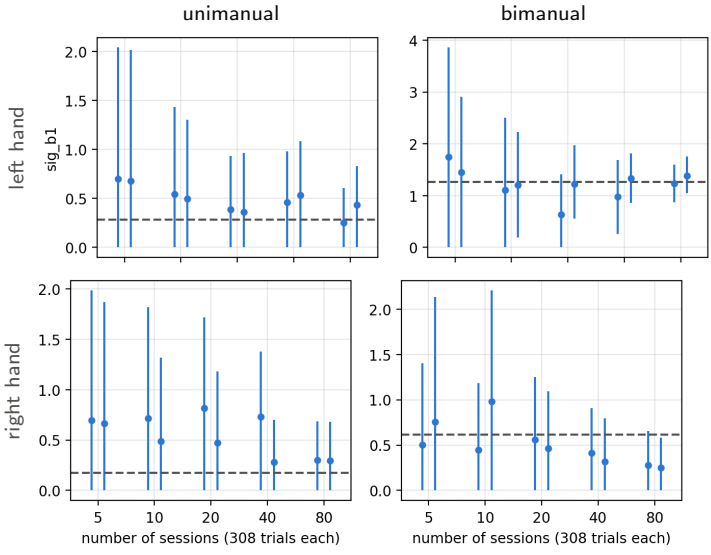
# Model B on synthetic data: $\mu_{\gamma_I}$ vs number of sessions



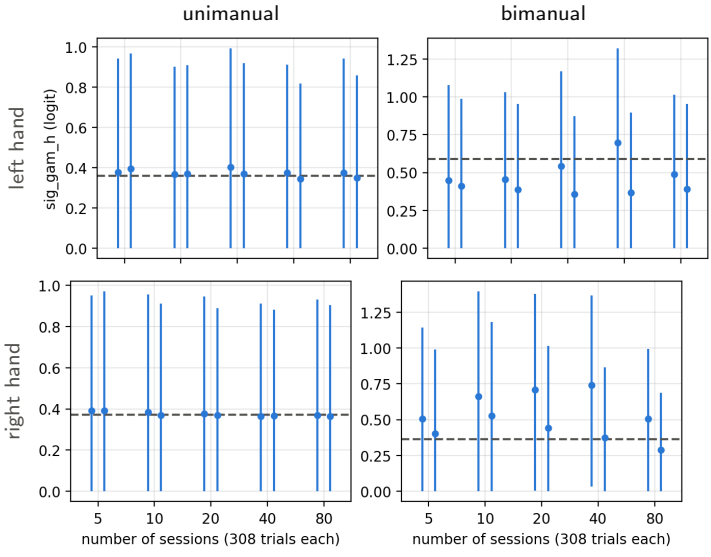
# Model B on synthetic data: $\sigma_{\beta_0}$ vs number of sessions



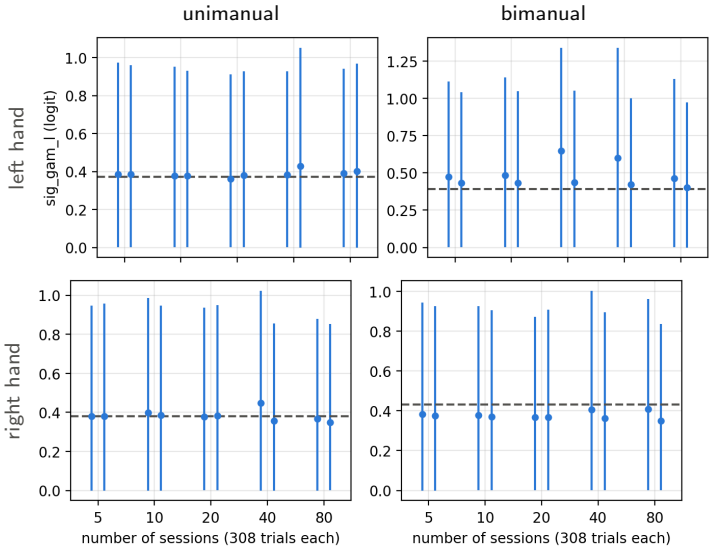
# Model B on synthetic data: $\sigma_{\beta_1}$ vs number of sessions



# Model B on synthetic data: $\sigma_{\gamma_h}$ vs number of sessions



# Model B on synthetic data: $\sigma_{\gamma_I}$ vs number of sessions



# Model B on synthetic data: session-level recovery

At 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean 95% HDI width / coverage

| | left unimanual | left bimanual | right unimanual | right bimanual |
|---|---|---|---|---|
| PSE | 0.08 / 0.33 / 0.95 | 0.11 / 0.42 / 0.98 | 0.11 / 0.38 / 0.92 | 0.11 / 0.40 / 0.92 |
| JND | 0.017 / 0.07 / 0.98 | 0.095 / 0.28 / 0.93 | 0.013 / 0.08 / 1.00 | 0.039 / 0.16 / 0.98 |
| $\beta_0$ | 0.45 / 1.83 / 0.96 | 0.40 / 1.47 / 0.92 | 0.49 / 1.76 / 0.90 | 0.46 / 1.75 / 0.92 |
| $\beta_1$ | 0.44 / 1.88 / 0.99 | 1.02 / 2.90 / 0.87 | 0.23 / 1.39 / 1.00 | 0.60 / 2.43 / 0.94 |
| $\gamma_h$ | 0.001 / 0.005 / 1.00 | 0.018 / 0.023 / 0.57 | 0.002 / 0.008 / 1.00 | 0.018 / 0.054 / 0.86 |
| $\gamma_l$ | 0.009 / 0.031 / 1.00 | 0.008 / 0.026 / 0.99 | 0.003 / 0.012 / 0.96 | 0.006 / 0.030 / 1.00 |

PSE and JND in standardized units (1 unit = 14.6 $\mu m$).

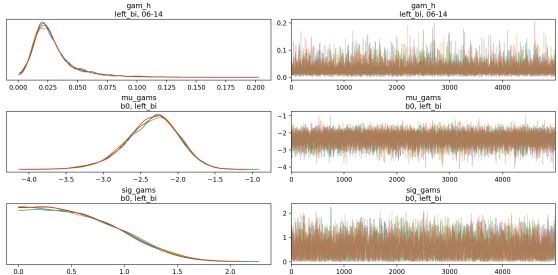
# Model B on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials), at values like the real ones:

- **Recoverable:** per-session PSE (RMSE about 1.5 $\mu m$) and JND; the group means $\mu_{\beta_0}$ and $\mu_{\beta_1}$; the bias spread $\sigma_{\beta_0}$; the slope spread $\sigma_{\beta_1}$ when it is large (left bimanual)
- **Weakly determined:** the lapse-rate means, especially $\mu_{\gamma_h}$ for unimanual trials ($\gamma_h$ near 0); smaller slope spreads, which cannot be told from 0
- **Biased in some datasets:** $\mu_{\gamma_l}$ for right unimanual (too high); $\gamma_h$ for left bimanual (too low) and $\gamma_l$ for left unimanual (too high), where the curve does not reach that asymptote
- **Overconfident:** session lapse rates. A miss in the group's $\gamma_h$ carries over to every session: bimanual session $\gamma_h$ HDIs cover the truth in only 57% (left) and 86% (right) of sessions
- **Not identified:** $\sigma_{\gamma_h}$, $\sigma_{\gamma_l}$; their posteriors reflect the prior

# Model B on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (31 min)
- No divergences; tree depth never above 6
- All but 4 of the 2,080 parameters: $\hat{R} = 1.00$, bulk ESS $\geq$ 2,210, tail ESS $\geq$ 4,527
- The 4 exceptions (raw `z_sig_b0`, $\hat{R}$ up to 1.73) are harmless: $\sigma_{\beta_0} = |2z|$, so chains differ only in the sign of $z$; $\sigma_{\beta_0}$ itself has $\hat{R} = 1.00$
- Below: left bimanual $\gamma_h$ (06-14) and its hyperparameters; $\sigma_{\gamma_h}$ piles up at 0 but mixes well



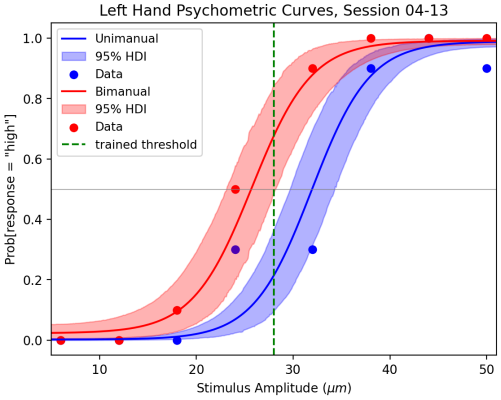
# Model B on the real data: a typical session

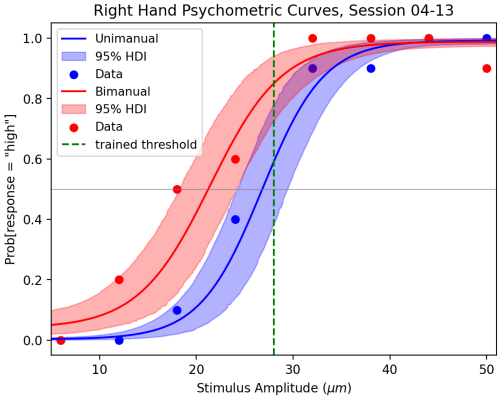
| | PSE ($\mu m$) | JND ($\mu m$) | $\gamma_h$ | $\gamma_l$ |
|---|---|---|---|---|
| left unimanual | 32.4 [31.6, 33.3] | 3.29 [2.95, 3.63] | 0.002 [0.000, 0.005] | 0.009 [0.000, 0.019] |
| left bimanual | 28.2 [27.2, 29.2] | 3.89 [3.34, 4.46] | 0.023 [0.009, 0.037] | 0.009 [0.001, 0.017] |
| right unimanual | 31.4 [30.3, 32.5] | 3.56 [3.23, 3.89] | 0.003 [0.000, 0.006] | 0.007 [0.000, 0.015] |
| right bimanual | 25.5 [24.5, 26.5] | 3.95 [3.48, 4.44] | 0.036 [0.017, 0.056] | 0.010 [0.003, 0.018] |

Quantities at the group-level means, posterior mean [95% HDI].

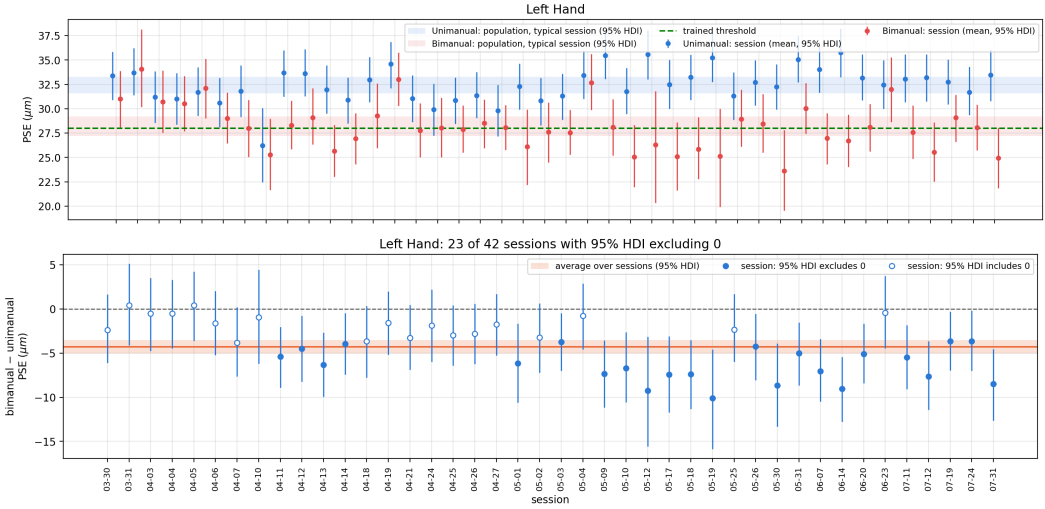
- The sessions differ in bias ($\sigma_{\beta_0} = 0.76$–$1.01$); left bimanual also in slope ($\sigma_{\beta_1} = 1.27$)
- The between-session lapse sds sit near their prior (not identified)

# Model B on the real data: fitted curves in one session (04-13)

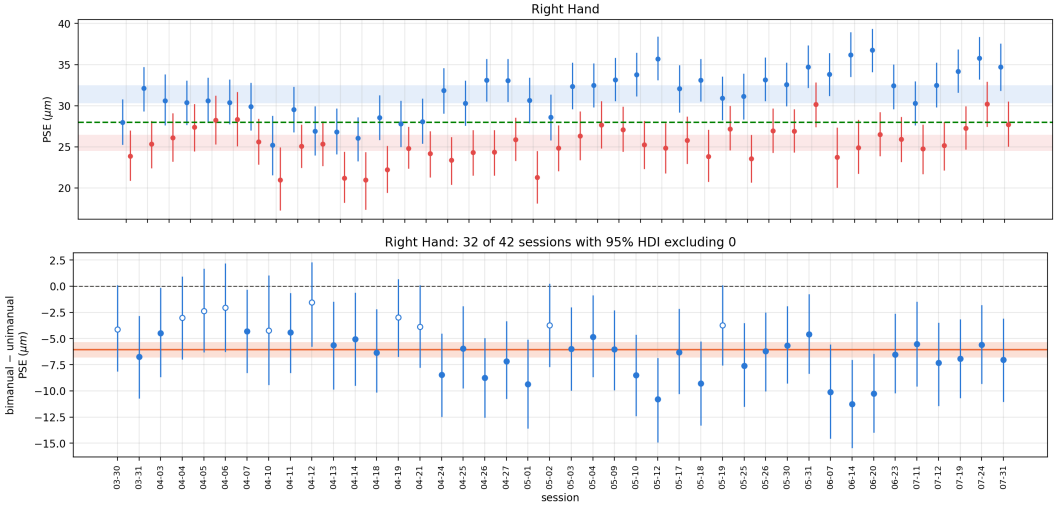


# Model B on the real data: PSE, left hand



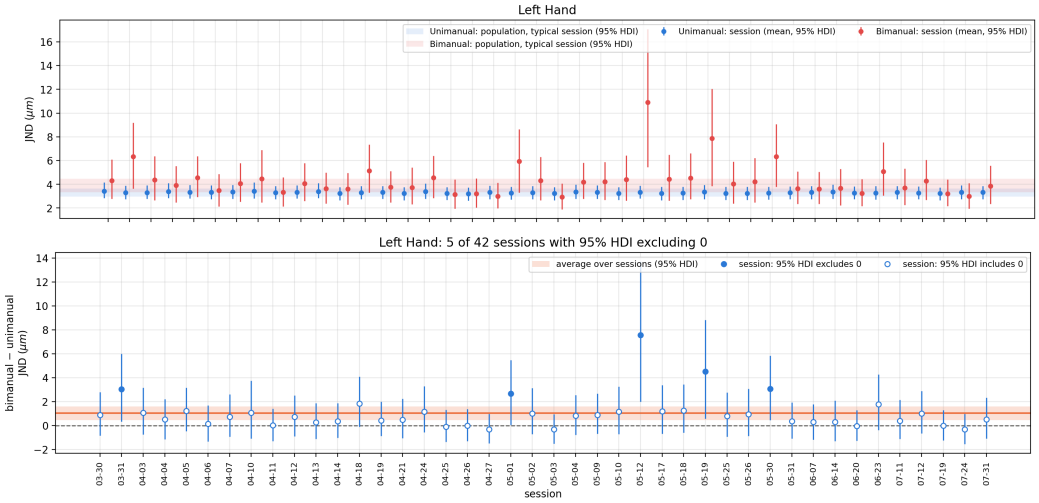
# Model B on the real data: PSE, right hand



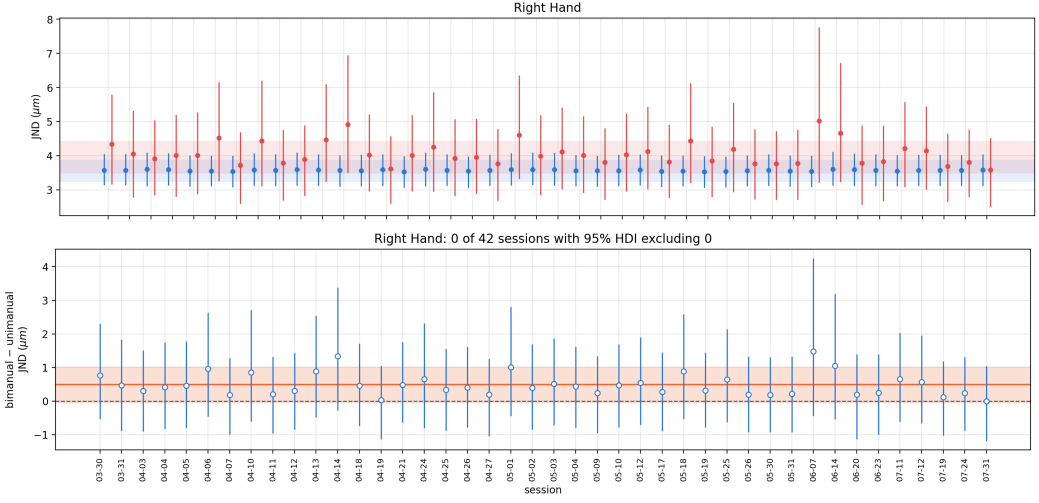

● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session; green dashed: trained threshold

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

# Model B on the real data: JND, left hand



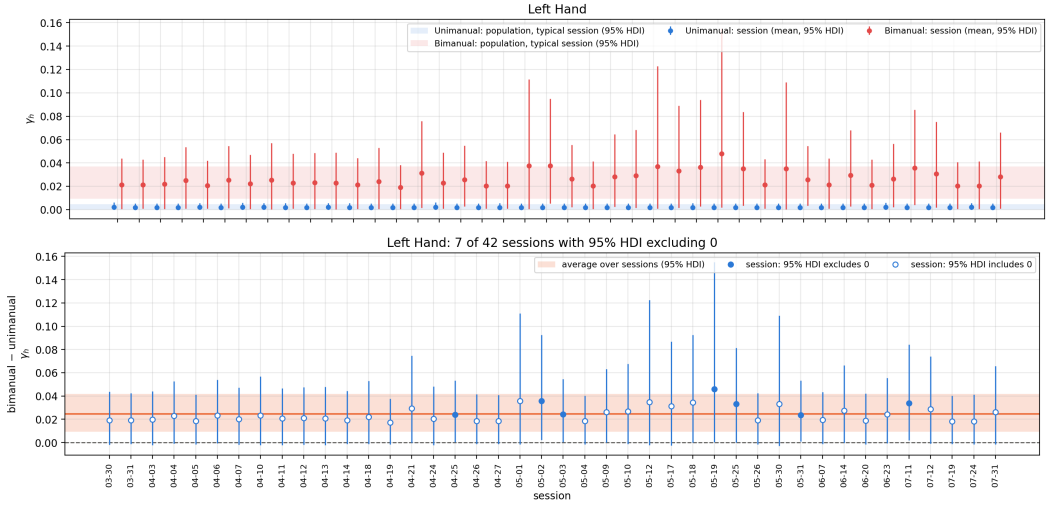
# Model B on the real data: JND, right hand



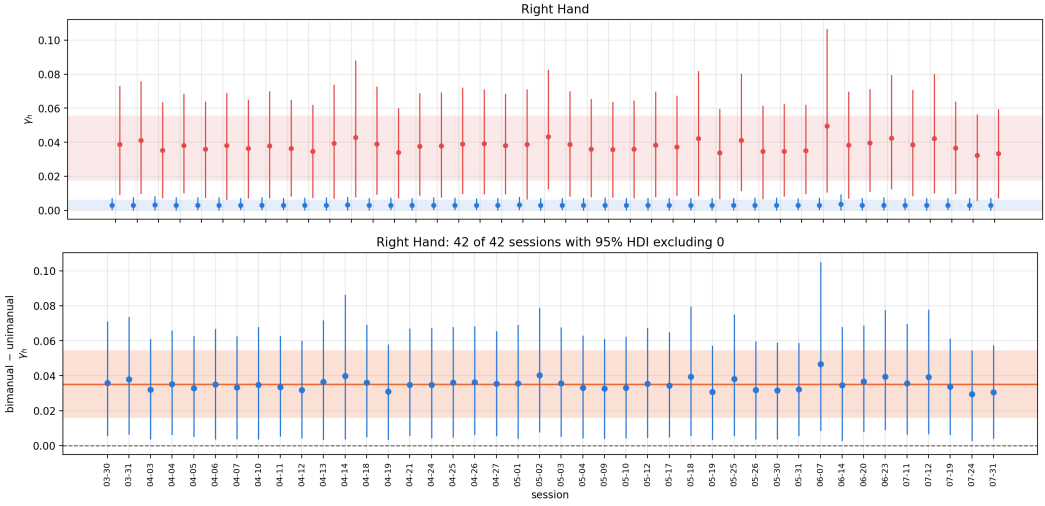

● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

# Model B on the real data: $\gamma_h$, left hand



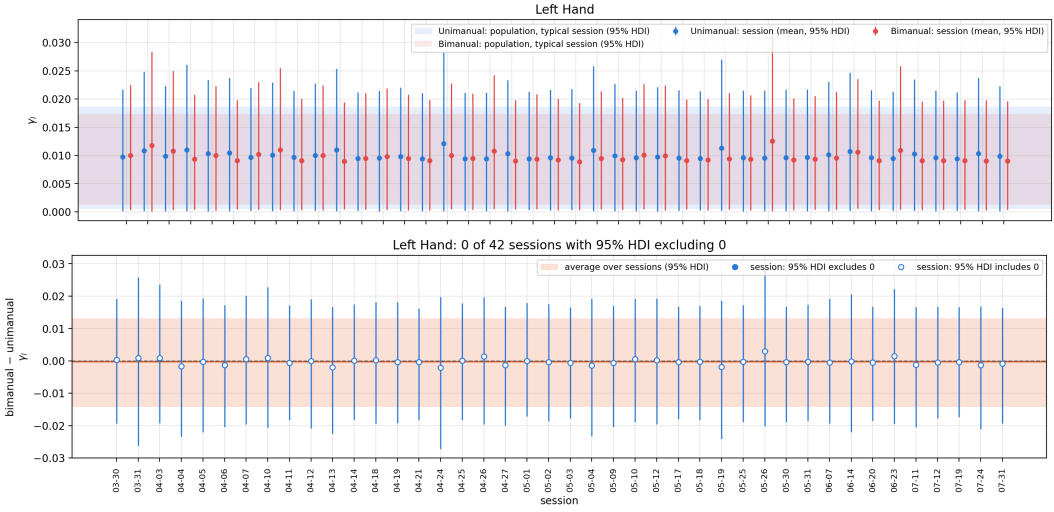
# Model B on the real data: $\gamma_h$, right hand



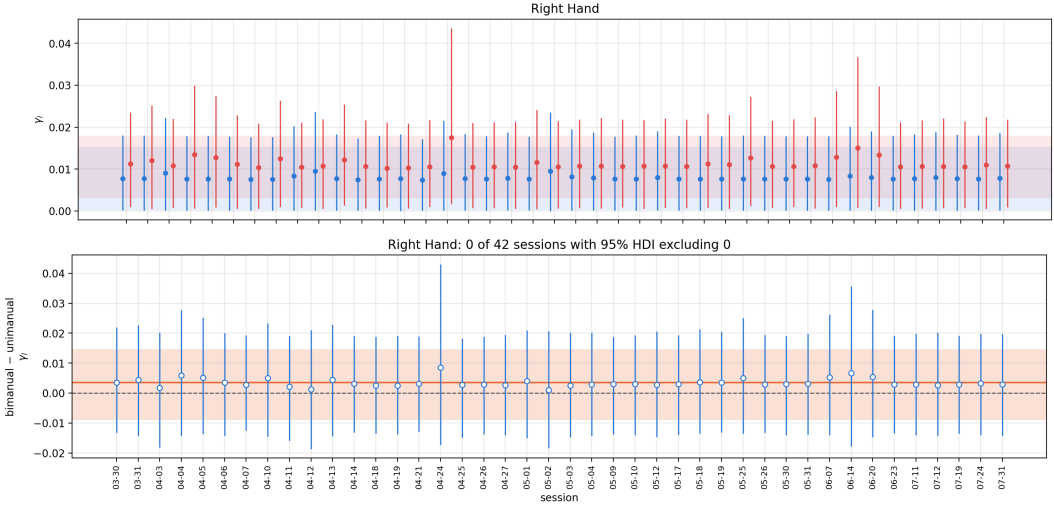

● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi — uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

# Model B on the real data: $\gamma_l$, left hand



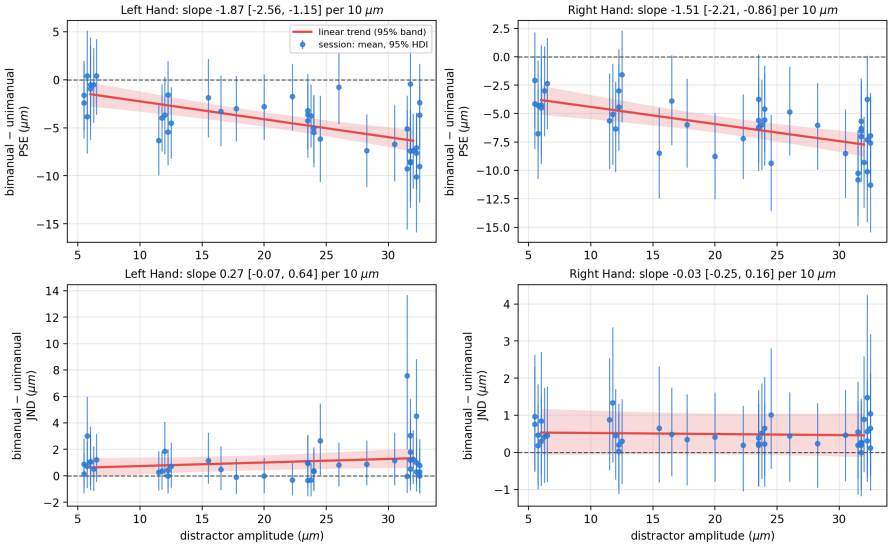
# Model B on the real data: $\gamma_l$, right hand



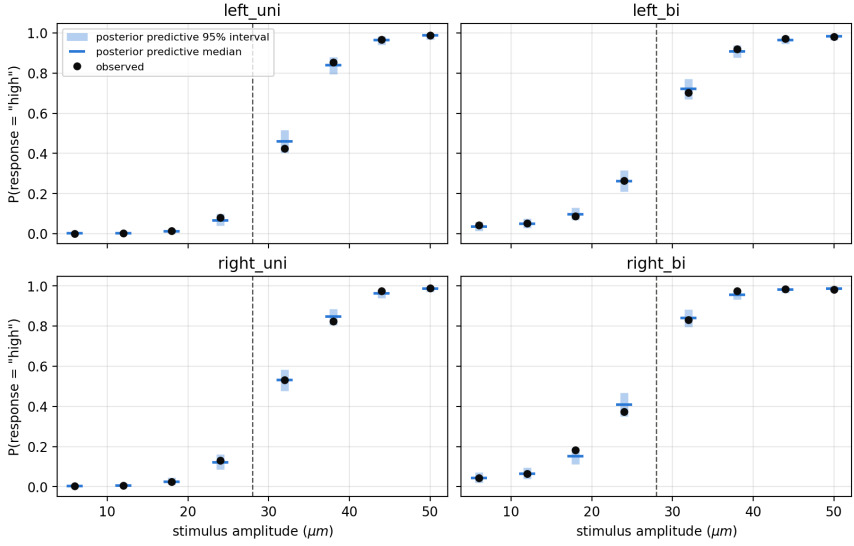

● unimanual, ● bimanual: session mean and 95% HDI; bands: typical session

bi − uni: filled if the session's 95% HDI excludes 0; orange band: average over sessions

# Model B: PSE shift vs distractor amplitude (exploratory)



# Model B on the real data: the pooled data are reproduced



# Model B: conclusions

- Reliable fit; no systematic misfit, pooled or within sessions ($\text{elpd}_{\text{loo}} = -3180.0$, SE 64.3)
- Effect of the distractor (bimanual − unimanual), average over the 42 sessions, posterior mean [95% HDI]:
  - **PSE:** lowered, by $-4.29$ [$-5.05$, $-3.52$] $\mu m$ (left) and $-6.06$ [$-6.80$, $-5.34$] $\mu m$ (right); posterior mean negative in 40/42 (left) and 42/42 (right) sessions
  - **JND:** raised for the left hand, $+1.03$ [0.46, 1.59] $\mu m$, driven by a few sessions; unclear for the right, $+0.49$ [$-0.07$, 1.03] $\mu m$
  - $\gamma_h$**:** raised, by $+0.025$ [0.009, 0.042] (left) and $+0.035$ [0.016, 0.054] (right)
  - $\gamma_l$**:** no effect, 0.000 [$-0.014$, 0.013] (left) and $+0.003$ [$-0.009$, 0.015] (right)
- The PSE shift grows with distractor amplitude, by about $-2$ $\mu m$ per 10 $\mu m$ (exploratory; amplitude is confounded with session order)

$\Rightarrow$ Put the distractor amplitude into the model: Model C

# Model C: including the distractor as a covariate

Psychometric function:

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}}\right)$$

Suppose the distractor has an impact on the perceived stimulus as follows:

$$x_{\text{perc}} = x_{\text{stim}} + c\, x_{\text{dist}}$$

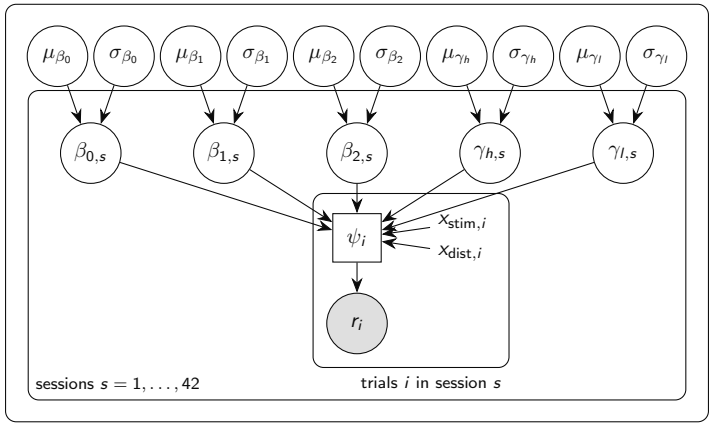
Then:

$$\begin{aligned}\beta_0 + \beta_1 x_{\text{perc}} &= \beta_0 + \beta_1(x_{\text{stim}} + c\, x_{\text{dist}}) \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_1 c\, x_{\text{dist}} \\ &= \beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\end{aligned}$$

Considering $x_{\text{dist}}$ as a covariate:

$$\psi(x_{\text{stim}}, x_{\text{dist}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}\left(\beta_0 + \beta_1 x_{\text{stim}} + \beta_2\, x_{\text{dist}}\right)$$

# Model C: graphical model




Circles: random variables (shaded: observed); rectangle: deterministic; $x_{\text{stim},i}$, $x_{\text{dist},i}$: fixed covariates. The same model is fit independently to each hand, with unimanual and bimanual trials together.

# Model C: likelihood

For each hand, fit independently to its $n \approx 6{,}470$ unimanual and bimanual trials, with the distractor as a covariate $x_{\text{dist},i}$ (amplitude / 14.63; $x_{\text{dist},i} = 0$ on unimanual trials):

Likelihood: probability of obtaining responses $\vec{r}$ given parameters $\vec{\theta} = (\vec{\theta}_s)_{s=1}^{42}$,
$\vec{\theta}_s = (\beta_{0,s}, \beta_{1,s}, \beta_{2,s}, \gamma_{h,s}, \gamma_{l,s})$
$r_i \sim \textbf{Bernoulli}[\psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})]$, with
$\psi(x_{\text{stim}}, x_{\text{dist}}; \vec{\theta}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}} + \beta_2 x_{\text{dist}})$, so:

$$\mathbf{P}[r_i = z] = \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})^z \left(1 - \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_{s[i]})\right)^{1-z}, \qquad z \in \{0, 1\}$$

$$\begin{aligned} L(\vec{\theta}; \vec{r}) &= \prod_{i=1}^{n} \mathbf{P}[r_i \mid \vec{\theta}_{s[i]}] \\ &= \prod_{s=1}^{42} \prod_{i:\, s[i]=s} \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_s)^{r_i} \left(1 - \psi(x_{\text{stim},i}, x_{\text{dist},i}; \vec{\theta}_s)\right)^{1-r_i} \end{aligned}$$

# Model C: priors

Session parameters drawn from the hand's population (non-centered in the code):

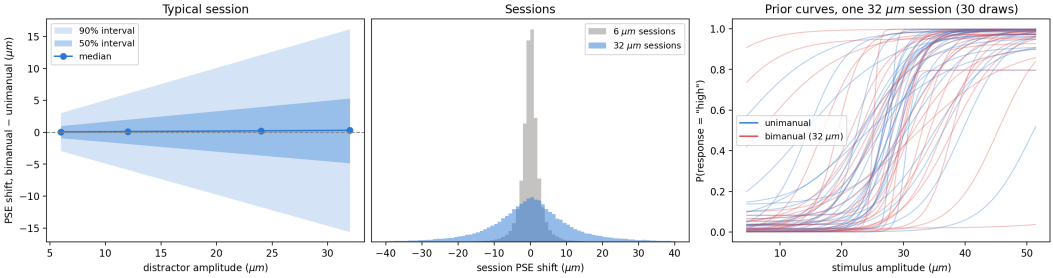
$$\beta_{k,s} \sim \mathcal{N}(\mu_{\beta_k},\ \sigma_{\beta_k}),\ k = 0, 1, 2, \quad \gamma_{h,s} = 0.25\, w_{h,s}, \quad w_{h,s} \sim \text{LogitNormal}(\mu_{\gamma_h},\ \sigma_{\gamma_h})$$
$$\gamma_{l,s} = 0.25\, w_{l,s}, \quad w_{l,s} \sim \text{LogitNormal}(\mu_{\gamma_l},\ \sigma_{\gamma_l})$$

Hyperpriors, the same for both hands; all but those of $\beta_2$ as in Model B:

$$\mu_{\beta_0} \sim \mathcal{N}(0,\ 2), \qquad \sigma_{\beta_0} \sim \text{HalfNormal}(2),$$
$$\mu_{\beta_1} \sim \text{LogNormal}(\log 4,\ 0.5), \qquad \sigma_{\beta_1} \sim \text{Exponential}(1),$$
$$\mu_{\beta_2} \sim \mathcal{N}(0,\ 1), \qquad \sigma_{\beta_2} \sim \text{HalfNormal}(1),$$
$$\mu_{\gamma_h},\ \mu_{\gamma_l} \sim \mathcal{N}(-3,\ 1.5), \qquad \sigma_{\gamma_h},\ \sigma_{\gamma_l} \sim \text{HalfNormal}(0.5)$$

- $\mu_{\beta_2}$ centered at 0: the distractor could push responses either way
- Uni- and bimanual trials share slope and lapse rates: $\beta_2 x_{\text{dist}}$ only shifts the curve

# Model C: prior predictive check of the distractor effect




**Model C drawback:** the PSE shift $-\beta_2 x_{\text{dist}}/\beta_1$ is tied to the distractor amplitude, so weak distractors can only shift the PSE slightly

# Model C on synthetic data: the study

- True hyperparameters derived from Model B's fit to the real data, so the synthetic data resemble the real data (e.g. $\mu_{\beta_2} = 0.93$ left, 1.13 right)
- Each simulated session gets one distractor amplitude, drawn from the real sessions' amplitudes
- The same two sweeps as for Model B, each run twice with new data:
  - sessions: 5, 10, 20, 40, 80 sessions of 308 trials
  - trials: 42 sessions of 20, 40, 80, 160, 320 trials
- 20 fits: 2 divergences in total (in the two smallest datasets), $\hat{R} \leq 1.008$, bulk ESS $\geq 532$

# Model C on synthetic data: hand-level means

At 40 sessions: true value and posterior mean [95% HDI], averaged over the two replicates

| | left | | right | |
|---|---|---|---|---|
| | true | posterior | true | posterior |
| $\mu_{\beta_0}$ | $-1.49$ | $-1.56$ [$-1.90$, $-1.24$] | $-1.06$ | $-0.94$ [$-1.29$, $-0.57$] |
| $\mu_{\beta_1}$ | 4.64 | 4.65 [4.16, 5.18] | 4.46 | 4.46 [4.06, 4.86] |
| $\mu_{\beta_2}$ | 0.93 | 0.93 [0.71, 1.18] | 1.13 | 1.11 [0.74, 1.47] |
| $\mu_{\gamma_h}$ | $-2.80$ | $-2.89$ [$-3.45$, $-2.41$] | $-2.41$ | $-2.60$ [$-3.10$, $-2.11$] |
| $\mu_{\gamma_l}$ | $-3.20$ | $-3.39$ [$-4.34$, $-2.60$] | $-3.20$ | $-3.04$ [$-3.73$, $-2.37$] |

- All five recovered without visible bias; $\mu_{\beta_2}$'s HDI excludes 0
- Lapse means (logit scale) are the least precise

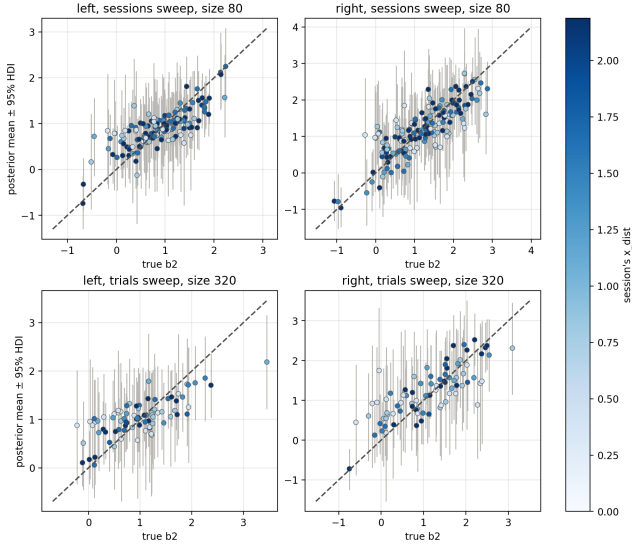
# Model C on synthetic data: between-session sds

At 80 sessions: true value and posterior mean [95% HDI], averaged over the two replicates

| | left | | right | |
|---|---|---|---|---|
| | true | posterior | true | posterior |
| $\sigma_{\beta_0}$ | 0.76 | 0.74 [0.58, 0.92] | 1.01 | 0.94 [0.76, 1.14] |
| $\sigma_{\beta_1}$ | 0.78 | 0.83 [0.57, 1.09] | 0.39 | 0.17 [0.00, 0.40] |
| $\sigma_{\beta_2}$ | 0.62 | 0.51 [0.36, 0.67] | 0.83 | 0.86 [0.68, 1.06] |
| $\sigma_{\gamma_h}$ | 0.48 | 0.38 [0.00, 0.83] | 0.37 | 0.31 [0.00, 0.68] |
| $\sigma_{\gamma_l}$ | 0.38 | 0.35 [0.00, 0.85] | 0.41 | 0.29 [0.00, 0.69] |

- The sds of $\beta_0$, $\beta_1$, $\beta_2$ are identified (a small $\sigma_{\beta_1}$ is hard to tell from 0)
- The lapse sds are not: their HDIs start at 0 and barely narrow with more data

# Model C on synthetic data: session $\beta_2$ is weak for small distractors



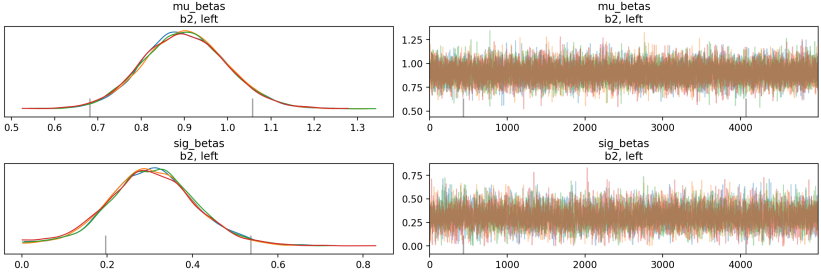
# Model C on synthetic data: what is recoverable

At the size of the real data (42 sessions of about 308 trials), if the data follow Model C:

- **Recoverable:** per-session PSEs in both conditions and JNDs; all five hand-level means, including the distractor effect $\mu_{\beta_2}$; the between-session sds of $\beta_0$, $\beta_1$ and $\beta_2$
- **Weakly determined:** the individual session coefficients, especially $\beta_2$ in sessions with a small distractor (HDI 2.6 wide at 6 $\mu m$, 1.2 at 32 $\mu m$), which are pulled toward the hand mean
- **Not identified:** $\sigma_{\gamma_h}$ and $\sigma_{\gamma_l}$; their posteriors reflect the prior
- **Not tested:** what happens when the data break Model C's assumptions (shared lapse rates and slope, a shift proportional to amplitude)

# Model C on the real data: sampling diagnostics

- NUTS, 4 chains $\times$ 5,000 draws after 1,000 tuning (9 min)
- 2 divergences in 20,000 draws, in no particular region; tree depth never above 5
- All 1,300 variables: $\hat{R} = 1.000$, bulk ESS $\geq$ 3,410, tail ESS $\geq$ 5,231
- HalfNormal sds defined directly, so no sign warning as in Model B
- Below: the left hand's distractor hyperparameters $\mu_{\beta_2}$, $\sigma_{\beta_2}$



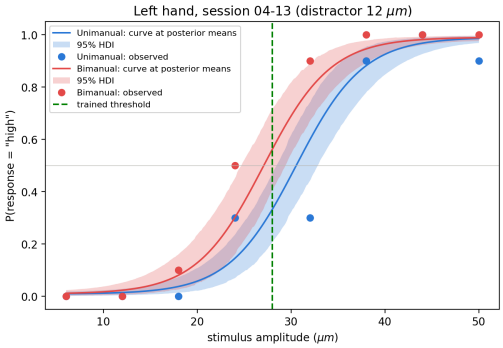
# Model C on the real data: hand-level estimates

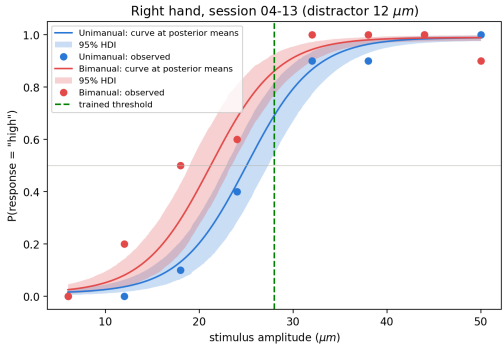
| | left | right |
|---|---|---|
| distractor coefficient $\mu_{\beta_2}$ | 0.90 [0.71, 1.09] | 1.18 [0.99, 1.37] |
| PSE, unimanual ($\mu m$) | 32.34 [31.51, 33.17] | 31.15 [29.92, 32.40] |
| JND ($\mu m$) | 3.75 [3.39, 4.11] | 3.88 [3.59, 4.19] |
| $\gamma_h$ | 0.0091 [0.0033, 0.0147] | 0.0113 [0.0042, 0.0189] |
| $\gamma_l$ | 0.0086 [0.0023, 0.0155] | 0.0085 [0.0029, 0.0143] |
| $\sigma_{\beta_0}$ / $\sigma_{\beta_1}$ / $\sigma_{\beta_2}$ | 0.60 / 0.90 / 0.30 | 1.07 / 0.35 / 0.23 |

Posterior mean [95% HDI]; PSE, JND and lapse rates of a typical session (JND and lapse rates are the same on bimanual trials).

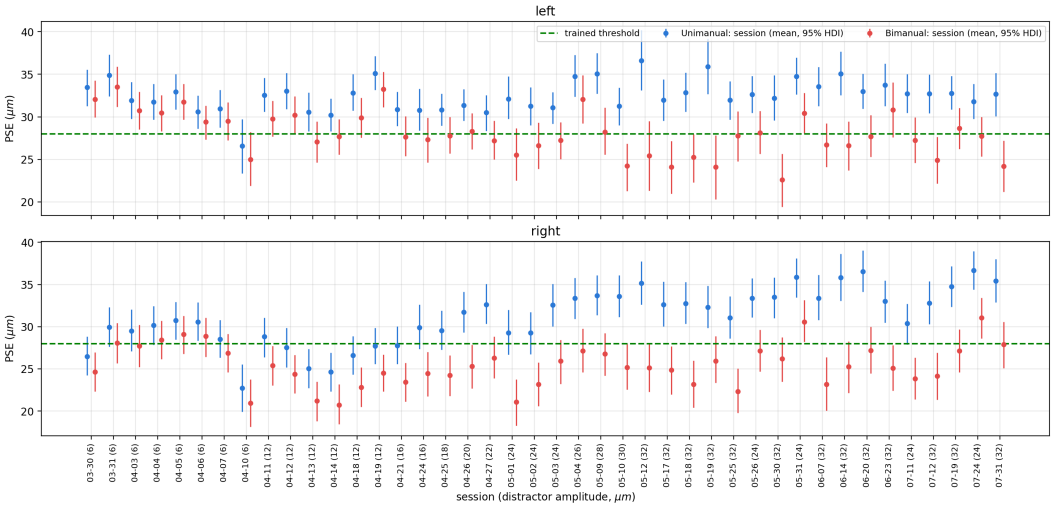
- $\mu_{\beta_2} > 0$ for both hands: the distractor lowers the PSE, more the stronger it is
- The sessions vary little around the hand's amplitude line ($\sigma_{\beta_2}$ small)

# Model C on the real data: fitted curves in one session (04-13)

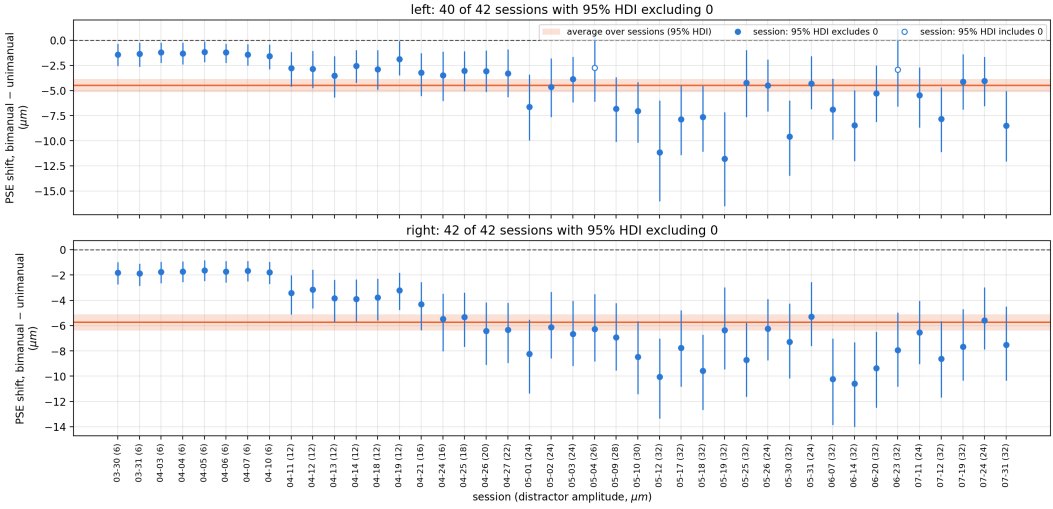


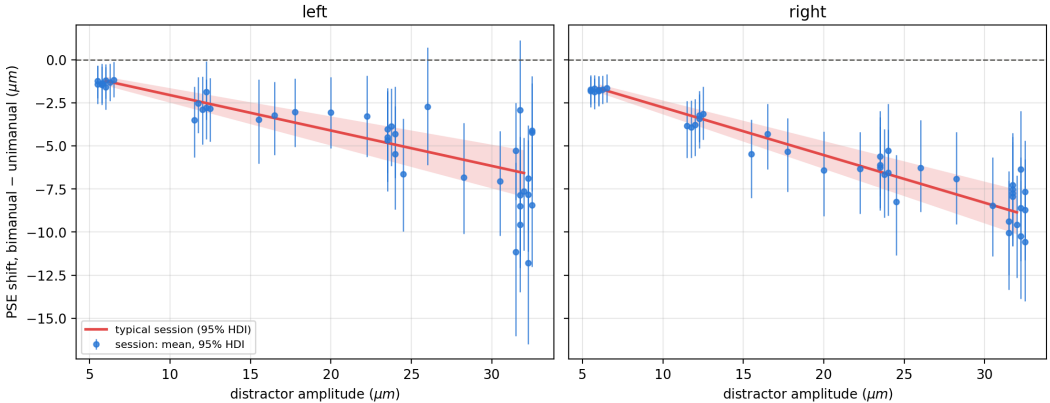
# Model C on the real data: PSE in each session



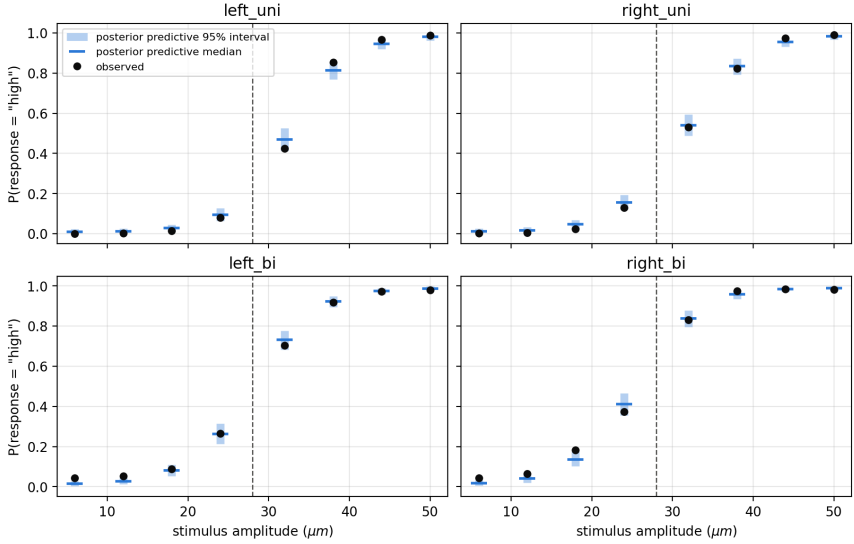
# Model C on the real data: effect of the distractor on PSE



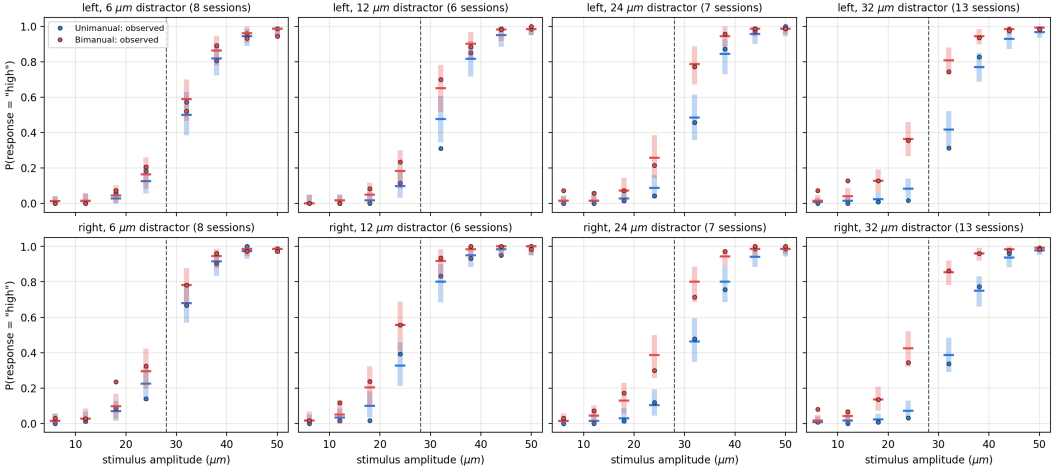
# Model C: the PSE shift grows with distractor amplitude



# Model C on the real data: misfit at the weakest stimuli



# Model C: observed vs predicted, by distractor amplitude



# Model C: conclusions

- Reliable fit: 2 divergences in 20,000 draws, $\hat{R} = 1.000$ ($\text{elpd}_{\text{loo}} = -3217.0$, SE 65.7)
- Effect of the distractor (bimanual − unimanual), posterior mean [95% HDI]:
  - **PSE:** lowered, by $-4.51$ [$-5.15$, $-3.86$] $\mu m$ (left) and $-5.74$ [$-6.39$, $-5.11$] $\mu m$ (right) on average over sessions; more the stronger the distractor: a typical session shifts $-1.23$ and $-1.66$ $\mu m$ at 6 $\mu m$, $-6.57$ and $-8.85$ $\mu m$ at 32 $\mu m$
  - **JND, $\gamma_h$, $\gamma_l$:** shared by both conditions, so the model cannot show an effect on them
- The misfit sits where its assumptions meet the data:
  - shared lapse rates: 6 of 32 pooled proportions outside their intervals, all at weak stimuli
  - a shift proportional to amplitude: for the right hand, even a 6 $\mu m$ distractor shifts responses more than Model C allows

$\Rightarrow$ Which model describes the data best? Model comparison

# Leave-one-out cross-validation

All three models fit to the same 12,937 trials in the same order, so the difference between two models can be computed trial by trial:

| Model | $\text{elpd}_{\text{loo}}$ (SE) | $p_{\text{loo}}$ | difference (SE) |
|---|---|---|---|
| A: sessions pooled | $-3344.0$ (64.5) | 13.4 | B $-$ A: $+164.1$ (19.2) |
| B: sessions hierarchical | $-3180.0$ (64.3) | 178.7 | **best** |
| C: distractor as covariate | $-3217.0$ (65.7) | 138.6 | C $-$ B: $-37.1$ (11.9) |

- B beats A by 8.5 standard errors, despite 179 effective parameters against 13
- B beats C by 3.1 standard errors, although C needs fewer effective parameters
- Every Pareto $k$ below 0.7, so the PSIS-LOO estimates are reliable

# Model recovery: is B's advantage real?

Two questions before reading B's win as session-to-session variation:

1. Can LOO identify the model that generated a dataset of this size and design?
2. Could B win just because it is more flexible? B contains A as the case of no between-session variation.

Design:

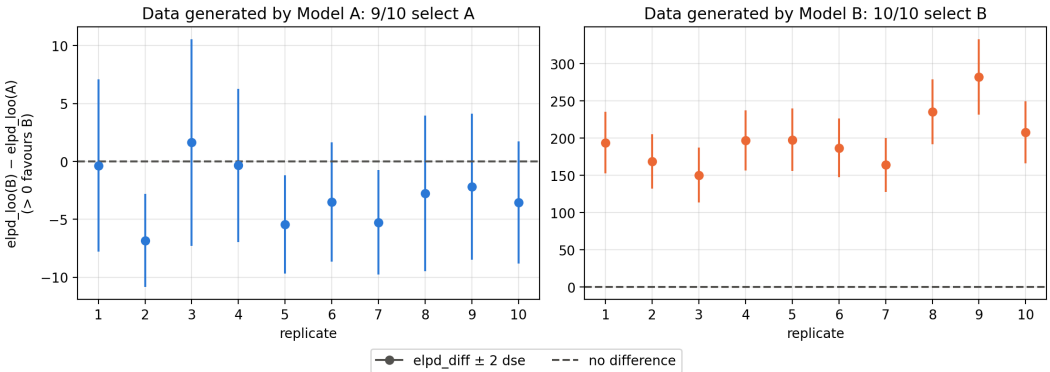
- 10 replicates, each with two synthetic datasets on the real trial design (12,937 trials in 42 sessions)
  - **data A:** responses from Model A, true values = A's real-data posterior means
  - **data B:** session parameters drawn from B's real-data hyperparameters
- Both models fit to every dataset; LOO picks the model with the higher $\text{elpd}_{\text{loo}}$; $\Delta = \text{elpd(B)} - \text{elpd(A)}$

# Model recovery: LOO picks the generating model

| | selects A | selects B |
|---|---|---|
| data from A | 9 | 1 |
| data from B | 0 | 10 |

- **Data from B:** B wins every time by a wide margin, $\Delta = 150.2$ to $282.4$ ($z = 8.2$ to $11.1$); B's $p_{\text{loo}} = 167$–$190$
- **Data from A:** A wins 9 of 10, but narrowly, $\Delta = -6.8$ to $+1.6$; the one B win is a near-tie ($z = 0.36$). B mimics A by shrinking its sds ($p_{\text{loo}} = 54$–$84$)
- **Real data:** $\Delta = 164.1$ and $p_{\text{loo}} = 178.7$, inside the data-from-B range and far outside the data-from-A range

# Model recovery: $\Delta$ in every replicate



# Model recovery: conclusions

- With this design and data size, LOO identifies session-to-session variation reliably: B wins every time with $z > 8$ when the variation is there
- B's flexibility alone does not make it win: on data without session variation it loses or ties ($\Delta \leq 1.6$)
- The real data look like the data-from-B replicates in both $\Delta$ and $p_{\text{loo}}$: the psychometric curves do vary between sessions
- The reverse, confirming that variation is absent, is only weakly answered: the more flexible model loses by just a few elpd points when the simpler one is true
- Limits: 10 replicates per model; one set of B hyperparameters (with smaller between-session sds the gap would shrink)

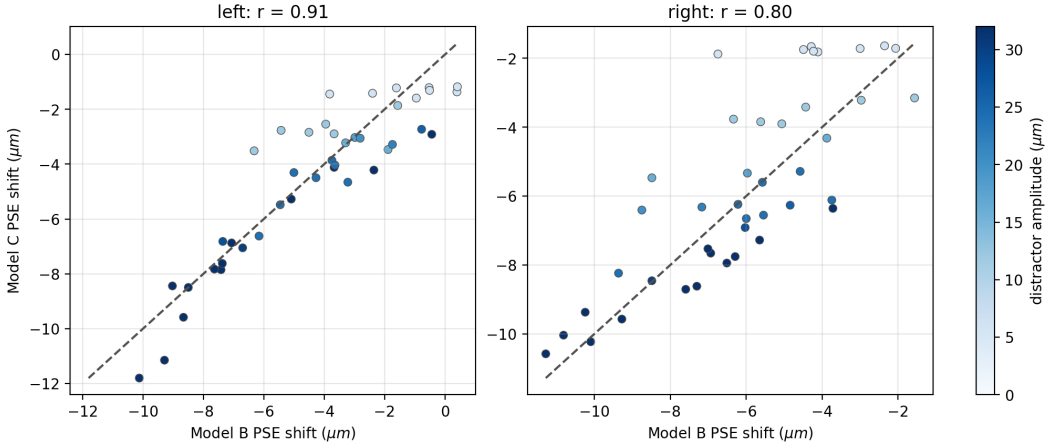
# Model C vs Model B: where C loses

$\text{elpd}_{\text{loo}}(\text{C}) - \text{elpd}_{\text{loo}}(\text{B})$ summed over the trials at each amplitude; negative values favor Model B

| | left | | right | |
|---|---|---|---|---|
| stimulus ($\mu m$) | bimanual | unimanual | bimanual | unimanual |
| 6 | −6.6 | −3.5 | −5.1 | −2.3 |
| 12 | −4.2 | −3.0 | −1.8 | −2.8 |
| 18 | 0.7 | −1.6 | −3.4 | −0.3 |
| 24 | −1.0 | 0.8 | 1.9 | 0.8 |
| 32 | −0.1 | −3.4 | 0.3 | 1.7 |
| 38 | −0.3 | −1.4 | 0.2 | 1.6 |
| 44 | 0.0 | −3.8 | 0.4 | −1.5 |
| 50 | −0.1 | 0.3 | 0.1 | 0.0 |

- 29.2 of the 37.1 are lost at 6 and 12 $\mu m$: one lapse rate for both conditions
- Next: left unimanual at 32 and 44 $\mu m$: one slope for both conditions

# Model C vs Model B: session PSE shifts



# Model C vs Model B: the weakest distractor

Session PSE shifts ($\mu m$): mean, and range of the posterior means, over the sessions at the weakest and strongest distractor

| hand | distractor | Model B | Model C |
|---|---|---|---|
| left | 6 $\mu m$ (8 sessions) | $-1.13$ ($-3.82$ to $0.41$) | $-1.34$ ($-1.59$ to $-1.18$) |
| left | 32 $\mu m$ (13 sessions) | $-6.67$ ($-10.12$ to $-0.44$) | $-7.40$ ($-11.80$ to $-2.91$) |
| right | 6 $\mu m$ (8 sessions) | $-3.91$ ($-6.74$ to $-2.06$) | $-1.75$ ($-1.88$ to $-1.64$) |
| right | 32 $\mu m$ (13 sessions) | $-7.90$ ($-11.28$ to $-3.72$) | $-8.58$ ($-10.58$ to $-6.36$) |

- The models agree on the average shift and, for the left hand, on the session shifts ($r = 0.91$)
- For the right hand at 6 $\mu m$, B's shift is more than twice C's: C ties the shift to the amplitude, so a weak distractor must give a small shift
- Suggests an effect of the distractor being present, on top of its amplitude

# Model comparison: conclusions

- **Sessions differ:** B predicts far better than A, and model recovery shows this is not B's flexibility alone.
- **The models agree on the PSE shift:** about $-4.2$ to $-4.5$ $\mu m$ (left) and $-5.7$ to $-6.1$ $\mu m$ (right) on average.
- **The shift depends on amplitude, but not only:** C confirms the trend inside a model, yet B still predicts better. C's misfit points to
  - lapse rates per condition (the strongest evidence)
  - a distractor-presence term, so the shift is not forced to 0 for a weak distractor
  - possibly a bimanual slope, which would let the distractor change the JND

$\Rightarrow$ Model B is the model to use for inference for now

# Comparison of the models

| | A | B | C |
|---|---|---|---|
| Sessions | pooled | hierarchical | hierarchical |
| Distractor | separate groups | separate groups | covariate $\beta_2 x_{\text{dist}}$ |
| $\text{elpd}_{\text{loo}}$ (SE) | $-3344.0$ (64.5) | $-3180.0$ (64.3) | $-3217.0$ (65.7) |
| difference from B (SE) | $-164.1$ (19.2) | best | $-37.1$ (11.9) |
| effective parameters $p_{\text{loo}}$ | 13.4 | 178.7 | 138.6 |
| PSE shift, left / right ($\mu m$) | $-4.19$ / $-6.07$ | $-4.29$ / $-6.06$ | $-4.51$ / $-5.74$ |

PSE shift: bimanual − unimanual, posterior mean; for B and C averaged over the 42 sessions. Numbers from Fit_A, Fit_B and Fit_C_Report.

- All three models agree on the PSE shift, to within about 0.3 $\mu m$ for each hand
- B predicts best: sessions differ (B $\gg$ A), and the shift is not simply proportional to distractor amplitude (B $>$ C)

# Conclusions

- The distractor lowers the PSE: by about 4.2–4.5 $\mu m$ (left) and 5.7–6.1 $\mu m$ (right), in all three models.
- It raises the rate of "high" responses to weak stimuli ($\gamma_h$).
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, but a weak distractor still shifts the PSE more than a proportional model allows (right hand).

# Next steps

- A new Model D, revising Model C: lapse rates per condition and a distractor-presence term.
- Model recovery for B against D.